

RSE3141 Solar Energy Assignment 3:

Solar PV and battery optimisation

Student Name: James Robinson

Student Number: 34988041

Word Count (not including references): 1963

How does the Model Work?

With the rise of domestic solar panel technology adjacent technologies are receiving more development. Batteries are increasing in popularity but are a large investment. An excel sheet by Chris Cooper and revised by Roger Dagarville is a complex system used to represent an average day of household energy supply and demand, simulating how much money a household could save annually by installing solar panels and a battery. The core logic is essentially dictated by a net power value. The Power Flow data sheet has two critical categories: solar generation and energy consumption, which are used to create this net power.

$$P_{net} = P_{generation} - P_{consumption}$$

The net power is calculated hourly and dictates the flow of energy. When P_{net} is positive there is a surplus of solar power which is directed to charge the battery, if the battery reaches maximum charge the surplus energy is directed to the grid as an export. If P_{net} is less than zero the consumption is higher than generation in this case the power drawn from the grid can be calculated as the difference between the magnitude of P_{net} and the power being discharged from the battery.

$$P_{grid} = |P_{net}| - P_{battery}$$

These calculations rely on the inputs of daily household energy usage, the general pattern for energy usage, (out of three options: stay at home, morning/evening peak, and controlled day loads), solar PV size, location specific yield factor (Melbourne =3.6 kWh/kW/day), battery capacity, round trip efficiency (92%), tariff structure (flat or time of use (ToU)), and export feed in tariff (FiT). These variables are manipulated to produce an output of annual savings. There is a feature in the code that forces a top up of the battery to 100% taking energy from the grid between 2 and 3 pm if not already full, this is because of lower ToU tariff prices during that time, ensuring when the highest tariff time comes the battery can be deployed to avoid paying the higher rates. Figure 1 depicts this change in tariffs showing a peak during the evening (4-9). Figure 2 shows the different energy usage types overlayed with the solar generation and peak tariff times. This is significant as the peak tariff period is after peak generation which is economically suboptimal hence the automated battery top up.

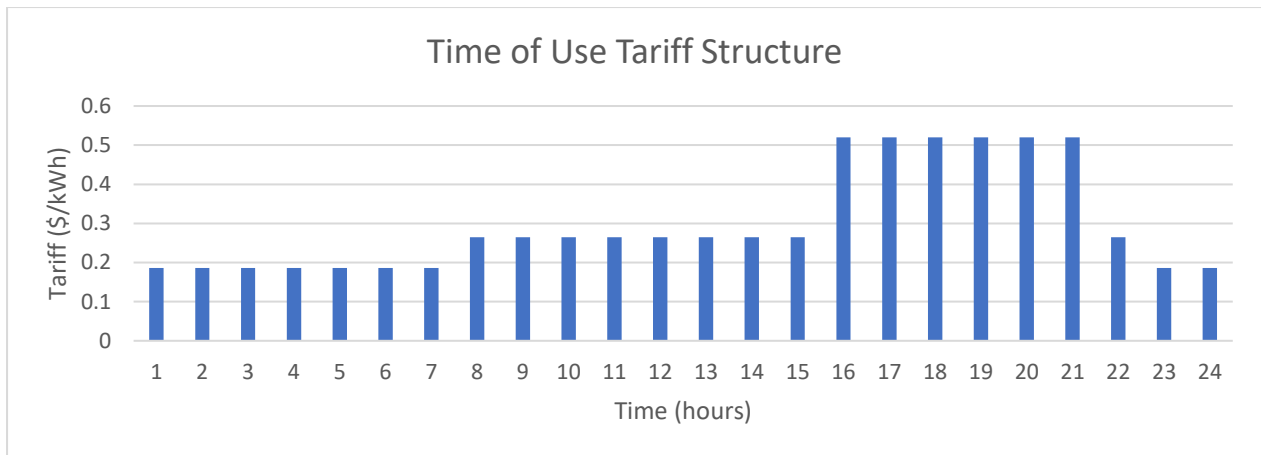


Figure 1: Shows how the time of use tariffs change throughout the day

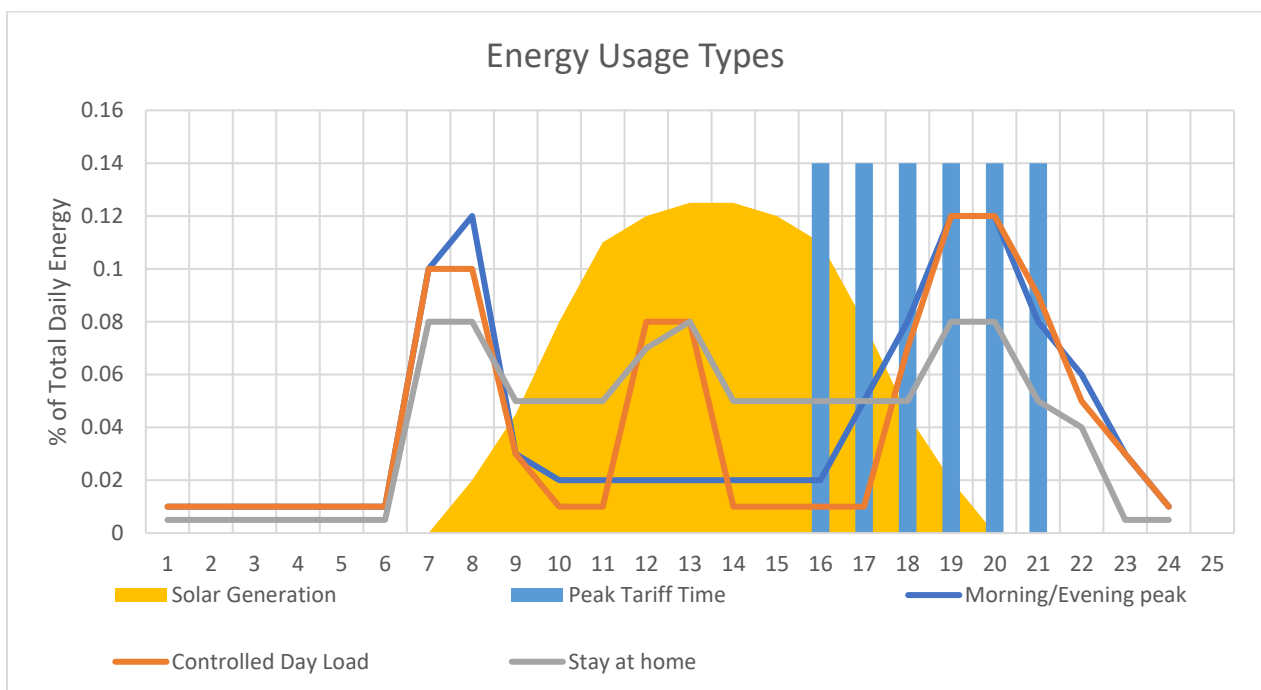


Figure 2: shows the energy demand profiles overlayed with solar production and peak tariff hours.

What are the Models Shortcomings?

In an effort to simplify the model and make it more accessible a number of compromises were made which jeopardise the accuracy of the model:

- **Average Solar day bias:** The calculations run on a single day of solar data to simulate the generation which serves as a rough estimate but ignores seasonality. It fails to represent long summer days with excess solar or colder winter days with less sun, this assumption creates an 'upper bound' bias, possibly overestimating realistic financial benefits.
- **Average Usage Day bias:** While three options are available to represent the usage throughout the day, again it is only on day that repeats. It fails to consider how household usage varies seasonally, such as running air conditioning in summer.

- **Lack of degradation:** The model assumes both the solar PV system and the battery output experiences no performance degradation over time, realistically lithium-ion batteries do suffer maximum capacity decreases of 1-3% annually (GridX, 2025) and solar panels lose efficiency of up to 0.55% annually (SolarQuotes, 2026) which decreases economic returns over time, particularly when the lifetime is set at 25 years for PV and 10 years for the battery.
- **Inverter size:** The model imposes a rigid 5kW cap on the inverter, automatically curtails energy flow to and from the battery that exceed this threshold. In real-world applications, inverter capacities are dynamically selected to match the specific solar array and battery specifications. By treating this as a fixed parameter, the model artificially restricts the system design window and may underestimate the economic value of higher-capacity battery configurations.

Modelling Scenarios

The report seeks to explore the economic viability of batteries in a PV system: at what cost is battery storage worth while and how does household contexts affect it? Different experiments were designed to test this with a baseline case identified by using solver to find the optimum PV capacity for common systems: no battery, 5kWh, 10kWh and 13.5kWh (Tesla Powerwall) taking the household energy usage as 15kWh/day following the morning/evening peak to represent a typical 9-5 working/kids at school household in Melbourne. Figure 3 compares this baseline showing increasing net profit with increasing battery capacity until the Tesla Powerwall at 13.5kWh where it actually decreases rapidly suggesting there is an upper limit of economically useful battery storage, which the larger battery has exceeded, hence it will be omitted from future experiments. Interestingly the recommended solar capacity is the same for no battery and a 5kWh battery: 2.6 kW, meaning the amount gained from exports must be around the same as saved by the battery. These baseline value optimal PV of and settings morning and evening peak, 15kW usage will be used in the future experiments to keep a comparable standard.

Scenario	Optimal PV (kW)	Annual Benefit (\$/year)	Costs (\$/year)	Net (\$/year)	Payback period (years)
PV only	2.6	548	204	344	4.8
PV + 5kWh	2.6	1460	680	780	4.2
PV + 10kWh	4.3	1946	1153	793	5.3
PV + 13.5kWh	4.6	2020.5	1411	610	6.1

Figure 3: The baseline economics based for different scenarios based on prior assumptions

Every household uses energy differently, so it is important to simulate how energy usage profiles impact benefits from a battery. The baseline uses morning and evening (M/E) peak usage, but this first experiment explores this impact of lifestyle on battery viability. Figure 4 compares the annual net benefit of each battery size for the different energy usage patterns. This experiment furthers the idea that there is diminishing returns for battery size as adding a 5kWh battery significantly increased earnings in all load types, yet 10kWh only minimally increased or actually decreased across the loading types. This experiment highlights that households who are out during the day have the most to benefit from adding a battery as it stores the energy for peak times when the occupants will be using the most energy and is at

its highest tariffs. Whereas households at home during the day gain less proportionally when compared to no battery as they have increased energy usage during the day which allows for the greatest overall returns, particularly with no battery installed, likely because the energy you generate is used, whereas in M/E the energy generated is wasted. This could be altered with electricity sent to the grid and earnings increased due to feed in tariffs, which are explored in the net experiment.

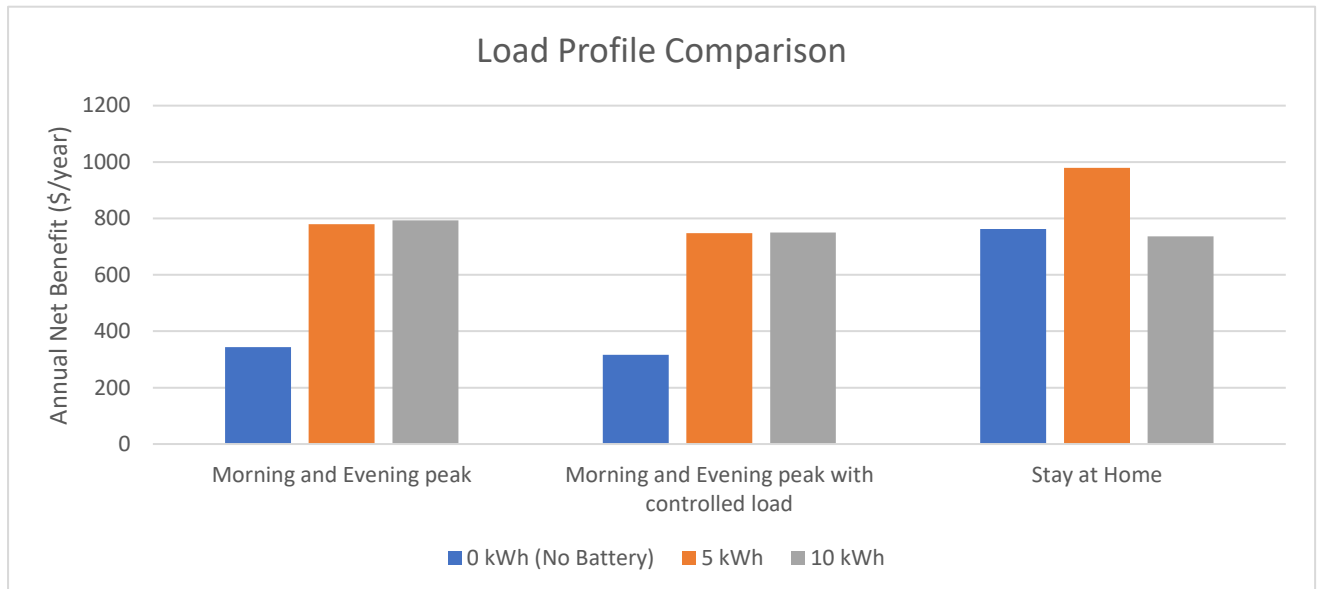


Figure 4: Net annual benefit for each demand profile in different battery scenarios

Excess energy is available to be sold to the grid; this experiment explores how feed in tariffs (FiT) impacts the battery viability. FiT vary greatly depending on the electricity provider with some offering higher rates during peak times, for this experiment values vary from 0\$/kWh to 0.12\$/kWh (Sykes, J 2026), again using M/E as they will benefit the most from FiT by not using energy during the day. Figure 5 has a very clear message, PV systems with a battery receive minimal increase in annual benefit as FiTs increases because less energy is being exported when compared to no battery, where almost all energy produced is sent to the grid. Hence no battery saw a significant change as FiTs increased, crucially even at the highest tariff rate in Victoria the benefits are still lower than having a battery.

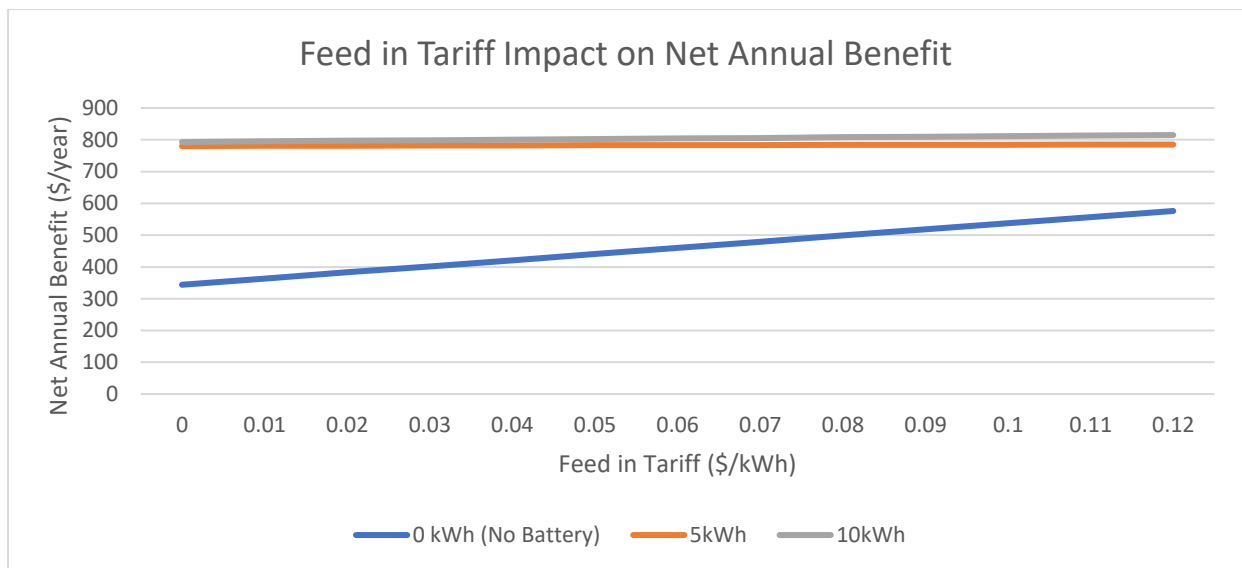


Figure 5: Annual benefits as Feed-in tariffs increase

The Australian government have a Solar Sharer Offer (SSO) where energy retailers offer households at least 3 hours of free electricity during the peak of solar generation in the middle of the day with Victoria introducing it in October. (Energy.gov.au, 2026), this significantly alters net annual benefits as seen in figure 6 where the data calculated with the SSO is lower than it's standard counterparts but follows the same trends. This decrease in savings is because when the household's solar panels are at peak generation the grid electricity is free thereby reducing the value of the energy generated during that time. However, it is worth noting the SSO estimates will be lower than the possible maximum savings as the sheet has an automatic battery top up from 2-3 (after SSO), if this were to be moved to during that period the battery would be topped up for free and increase annual benefits.

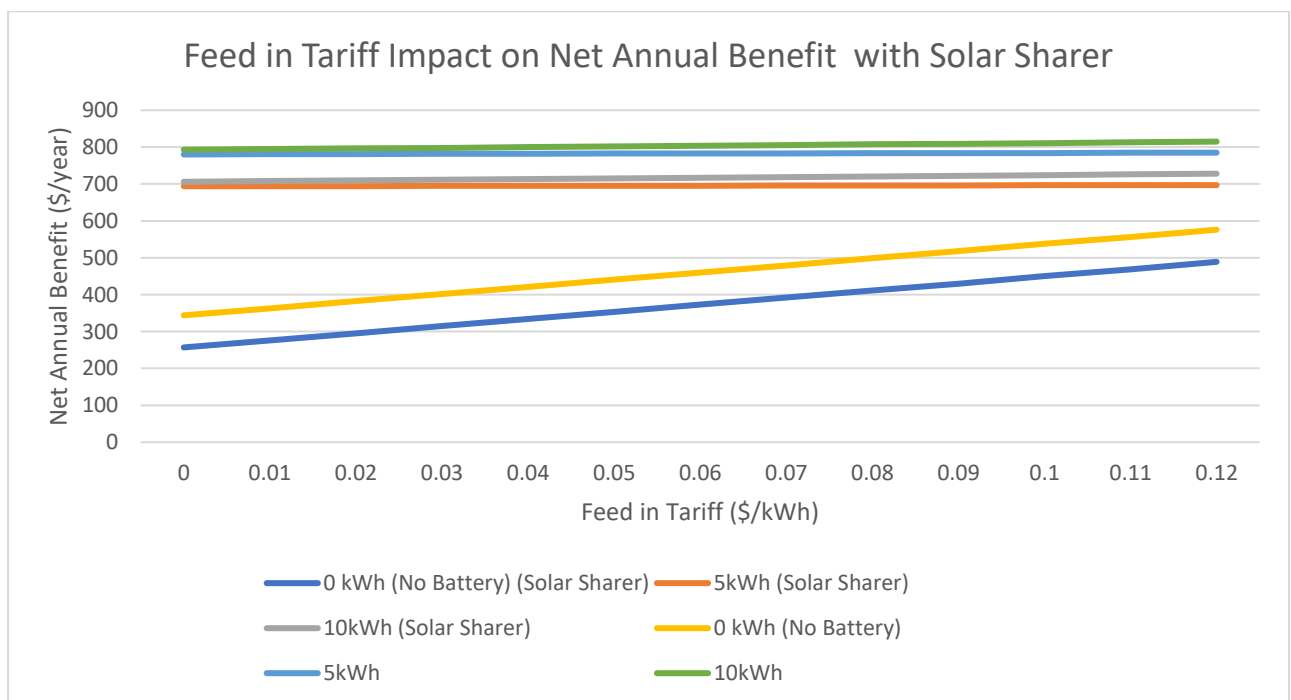


Figure 6: Annual benefits as Feed-in tariffs increase, with addition data calculated using the solar sharer offer tariff structure

The final experiment examines the impact that varying battery cost has on the net annual benefit. The cost is varied from 100 to 1000\$/kWh. Obviously the PV only setup was not affected by battery prices since there is no battery so that can serve as the viability line. In figure 7 the 5kWh battery sits above the PV only line the entire span suggesting that the smaller battery is cheaper and worth while having in all PV homes, where as the 10kWh battery sits below the PV only line beyond about 800\$/kWh suggesting it is worth investing if the cost is below that threshold. Interestingly at 500\$/kWh there is a shift from 5kWh being more economical to 10kWh suggesting it is worth upgrading to the larger battery as the costs decrease. Figure 8 shows there is a trend of prices decreasing which is indicative of the technology getting better and batteries becoming more affordable, increasing the viability. It is worth noting that the values in figure 8 are inclusive of installation costs which are relatively the same regardless of capacity, hence the lower per kWh costs for larger batteries.

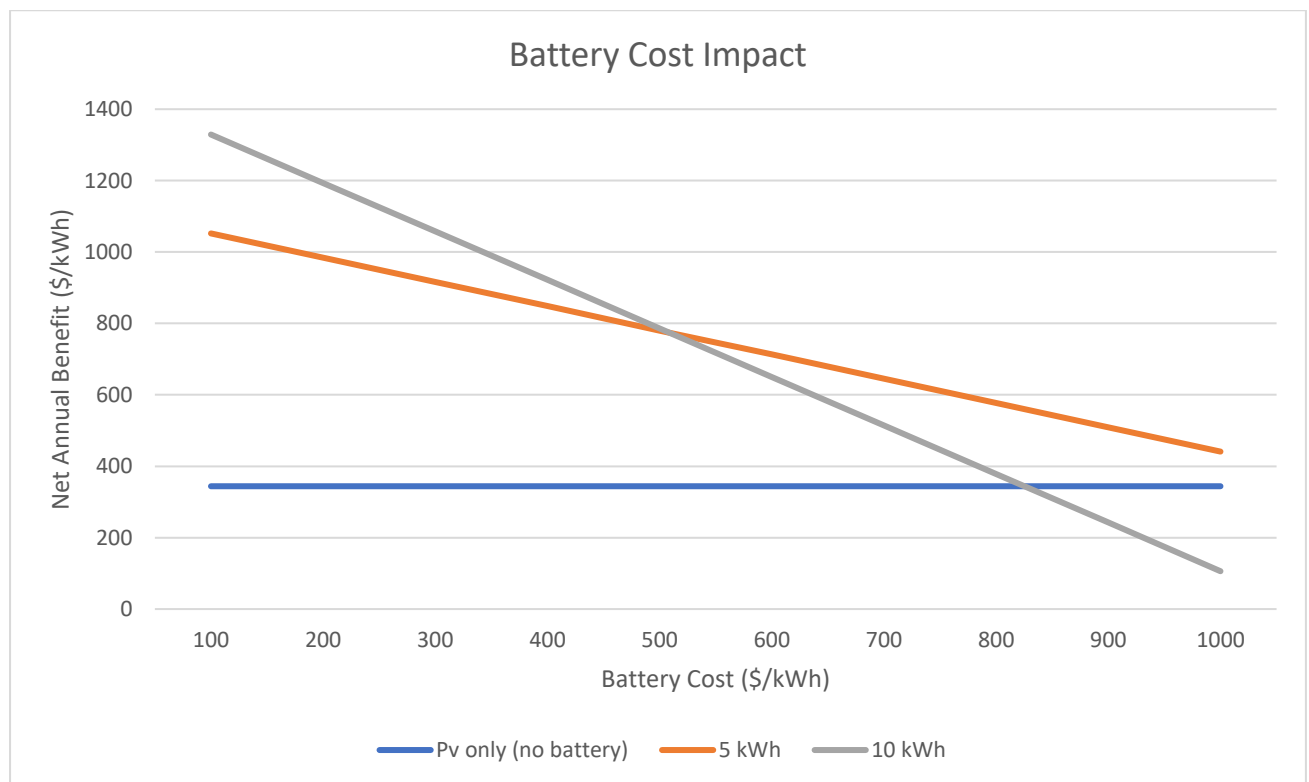


Figure 7: Net annual benefit of different systems as battery prices change

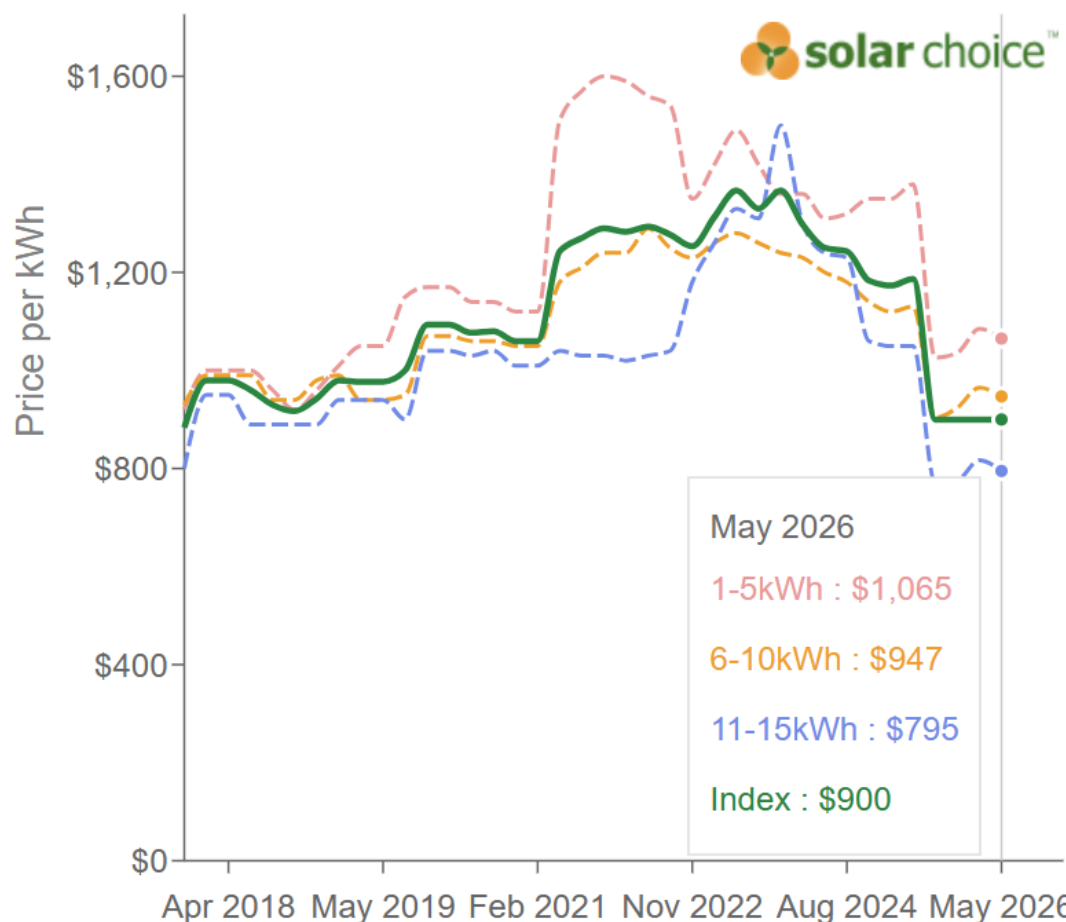


Figure 8: Current battery installation prices in Australia (Sykes, J. Solar Choice, 2026)

Discussion and conclusions

The experiments collectively build a picture of where battery storage technology sits economically in 2026 and how different conditions can shift that position. The baseline optimisation established PV alone delivers returns with batteries drastically increasing the benefit, but it is not linear. A 5kWh battery captures most of the available value while larger systems suffer from increased capital costs which detracts from the net benefit, evident as the 13.5kWh Powerwall exceeded this threshold and eroded the revenue.

The load profile experiment demonstrated that battery value is tied directly to the mismatch between solar generation and consumption timing. M/E households, representing a typical working family, benefit the most from installing storage because their demand is concentrated when solar generation is 0 and tariffs are highest. In contrast the stay at home profile already captures much of the solar energy through direct consumption leaving less opportunity for a battery to exploit. The implication is that battery recommendations should be tailored based on lifestyle rather than being applied uniformly.

The feed in tariff experiment revealed an important tension in solar policy. Higher FITs improve returns for PV only systems, thereby reducing the benefit of owning a battery since storing energy with high capital costs is less attractive when exporting it pays well. The Solar Sharer Offer (SSO) introduces a related dynamic, free midday grid electricity reduces value of solar

generation during those peak production hours, reducing savings across all configurations. However, if the battery were to top up when there is access to free electricity there would be more beneficial outcomes, reinforcing the limitations of this model's approach.

The battery cost experiment illustrates the growing benefit as battery prices decrease. It shows at current battery prices above 500\$/kWh a 5kWh system bares greater returns, however the current trajectory seen in the Solar Choices graph indicate a large decline from the 2023-2023 peak. If this trend continues larger batteries will offer larger benefits.

The analysis demonstrates PV remains the primary option to maximise values as it has lower capital costs than batteries which offer incremental yet marginal returns in many cases for households in 2026. The economics of battery storage a sensitive to three main variables, experimented in: demand profile, FiTs, and battery costs. However falling prices and improving technologies increase the likelihood for batteries to become far more mainstream. Government policies such as the SSO shift the economics as the world adapts to increased renewable energy reliance so the field is constantly evolving. Despite the shortcomings of the model, it offers insights to the intricacies of battery economics.

References

- Brakels, R. (2026). How Long Do Solar Panels Last? Degradation Rates in 2026 Compared. Available at: <https://www.solarquotes.com.au/panels/degradation/>
- Energy.gov.au. (2026). Solar Sharer Offer to cut electricity bills. Available at: <https://www.energy.gov.au/news/solar-sharer-offer-cut-electricity-bills>
- GridX. (2025). Battery degradation. Available at: <https://www.gridx.ai/knowledge/what-is-battery-degradation-and-how-to-prevent-it>
- Sykes, J. (2026). Best VIC Solar Feed-in Tariffss (2026). Available at: <https://www.solarchoice.net.au/learn/energy/feed-in-tariffs/best-vic-fit/>
- Sykes, J. (2026). Solar Choice Battery Price Index. Available at: https://www.solarchoice.net.au/solar-batteries/price/?sca=goog_ad_pmax&gad_source=1&gad_campaignid=20389884718&gbraid=0AAAAABhSmkniiYU9Wa15xd8xHMDGaqzNa&gclid=Cj0KCQjwoMXQBhDcARIsAH-eEtvRoRuc2XL7U9yh_Duh12T0ETr5glZPLDt6lVdsthzwD8SqwbxLLZQaAjq0EALw_wcB